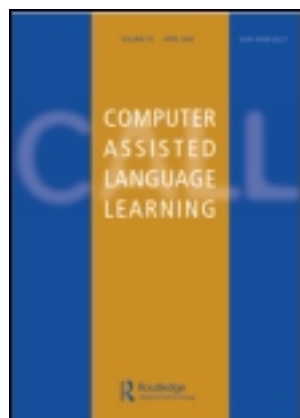


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Technologies for foreign language learning: a review of technology types and their effectiveness

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Technologies for foreign language learning: a review of technology types and their effectiveness

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This review summarizes evidence for the effectiveness of technology use in foreign language (FL) learning and teaching, with a focus on empirical studies that compare the use of newer technologies with more traditional methods or materials. The review of over 350 studies (including classroom-based technologies, individual study tools, network-based social computing, and mobile and portable devices) revealed that, in spite of an abundance of publications available on the topic of technology use in FL learning and teaching, evidence of efficacy is limited. However, strong support for the claim that technology made a measurable impact in FL learning came from studies on computer-assisted pronunciation training, in particular, automatic speech recognition (ASR). These studies demonstrated that ASR can facilitate the improvement of pronunciation and can provide feedback effectively. Additional studies provided strong support for the use of chat in FL learning. These studies showed that, with chat, both the amount of learners' language production and its complexity significantly increased. The literature revealed moderate support for claims that technology enhanced learners' output and interaction, affect and motivation, feedback, and metalinguistic knowledge.

Keywords: CALL; foreign language; technology; mobile; Web 2.0

Introduction

Well-established technologies, such as the personal computer and internet access, have become nearly ubiquitous for foreign language (FL) learning in many industrialized countries. In addition, relatively new technologies, such as smartphones and other mobile internet-accessible devices, are increasingly available. Other technologies, such as natural language processing (NLP), are still maturing. As technologies mature, become readily available, and are adapted for FL pedagogy, instructors may alter their teaching strategies or adjust their teaching activities to most effectively utilize available resources. At their best, technological innovations can increase learner interest and motivation; provide students with increased access to target language (TL) input, interaction opportunities, and feedback; and provide

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instructors with an efficient means for organizing course content and interacting with multiple students. At their worst, the use of new technologies can result in inappropriate input, shallow interaction, and inaccurate feedback; student frustration with software and hardware; distraction from the learning task; and a general over-emphasis on delivery modality over learning objectives.

Computer-assisted language learning (CALL) has been a subject of investigation for over 30 years, yet the state of the CALL literature leaves much to be desired in terms of a unified research agenda (Chappelle, 1997; Felix, 2005) and durable, validated findings (Felix, 2005; Stockwell, 2007a). Common problems in CALL research include: poor description of the research design; poor choice of variables to be investigated; lack of relevant data about participants; studies based on untrained users of the technology; a nearly exclusive focus on Western European languages, especially English; and an overall lack of systematicity in investigating key factors that may enhance the effectiveness of FL learning (Felix, 2005; Hubbard, 2005; Stockwell, 2007; Zhao, 2003). This situation creates a challenge for FL pedagogues and Second Language Acquisition (SLA) researchers attempting to evaluate the efficacy of technology use in FL learning and teaching.

Focus and scope of this review

This review summarizes evidence for the effectiveness of technology use in FL learning and teaching. Therefore, we focus on empirical studies that include research-based evidence showing that technology was effective and not on descriptions of technologies or their potential uses. Table 1 provides a list of technologies reviewed in this report and their brief descriptions.

We have specifically excluded the personal computer (desktop and laptop PC) and internet connectivity. Their near-total, if not total, use by faculty and students in post-secondary education speaks for itself; at least, in terms of acceptance and adoption, if not effectiveness for instruction and learning. Similarly, we did not include existing technologies that have been available for use in teaching for at least a few decades (e.g. televisions, videotapes, audiotapes) or those whose functionalities primarily provide minor modifications to previous similar technologies (e.g. DVD and CD players; digital slide presentation hardware or software); CDs or audiotapes; standard e-mail; or digital slide presentation technologies. The technologies selected for review do not cover the entire space of possible technological aids to FL teaching. In addition to excluding those that we felt were well-established both in society and in the classroom, we also excluded those that we judged as still needing to mature to be widely adapted for CALL. For instance, we focused only on those subfields of NLP that are closely related to CALL (e.g. ITS, ASR, corpus linguistics); however, we recognize that as the other subfields of NLP mature, they will make their way into CALL and NLP will become a fully recognized research area within CALL.¹ We further chose to limit this review to studies that focus on a single application of technology instead of multiple technologies at once or CALL in general. However, where applicable, we do include meta-analyses if they include studies on single applications. Over 350 potentially relevant publications were identified using these constraints and the subset described in this article was selected because these studies met our earlier established criteria for empirical evidence.

Table 1. Reviewed technologies, their brief descriptions, and some affordances of each.

		Example affordances for language study
Schoolhouse- or classroom-based technologies		
Course management system (CMS)	Server-based application used to present materials and services required for blended or distance learning (such as syllabi, required readings, calendars, etc.). Teachers and students access a CMS over a network through a web browser, using a menu-driven interface	• Enable sharing of course materials, allowing accessibility to content anytime, anywhere • Facilitate course content organization and teacher–student and student–student communication
Interactive white board	An interactive display that comprises three pieces of equipment: a computer, a projector, and a display panel, which is a large free-standing or wall-mounted touch-sensitive screen. The projector displays the image of the computer screen on the screen, which is easily viewable by all students in the classroom	• Promote interactive activities and engage students and teachers in collaborative work • Enhance motivation and improve attitudes toward learning • Incorporate authentic content available on the internet into classroom lessons
ePortfolio	A digital archive of student work created by a learner that records evidence of the learner's experiences, progress, achievements, and self-reflections	• Support learner autonomy and self-assessment • Emphasize the process of learning, rather than just the products of learning • Facilitate setting learning goals, monitoring progress, and developing self-assessment skills (Blackburn & Hakel, 2006)
Individual study tools		
Corpus	A collection of authentic language in spoken form, written form, or both. Corpora vary in terms of design (fixed size vs. expandable), content (general vs. specialized), and medium (written vs. spoken)	• Provide access to rich, authentic input • Enable broad access to linguistic data • Promote data-driven inductive learning (Bernardini, 2004; Leech, 1997)
Electronic dictionary	A dictionary in electronic form – either handheld or online	• Speed searches for a lexical item so that looking up words does not greatly interrupt the reading process

(continued)

Table 1. (Continued).

		Example affordances for language study
Electronic gloss or annotation	A method of reference, usually in a form of a hyperlink, that allows learners to access glosses (word- or sentence-level, context-specific translations) or annotations (explanatory or background information) while reading an electronic text	● Accommodate different look-up preferences and learning styles ● Support individualized and elaborated input ● Provide for efficient look-up of unknown words and multimedia capability ● Facilitate reading comprehension, and incidental and intentional vocabulary learning
Intelligent tutoring system	A program that simulates a tutor by providing direct, customized instruction and/or feedback to a learner. Such a system is generally comprised of four components: an interface (platform), an expert model (domain of knowledge the student is intended to acquire), a student model (current state of student's knowledge), and a tutor model (which provides appropriate feedback and instruction by using the identified gaps between the student and the expert models)	● Tailor instruction to the individual learner ● Provide immediate, specific feedback in a systematic manner ● Can implement task-based interfaces in language instruction
Grammar checker	A program designed to evaluate a written text's well-formedness in terms of grammaticality. Such programs are often packaged, along with spellcheckers, within word-processing programs	● Identify/flag low-level morphosyntactic errors (Burston, 2001; Jacobs & Rogers, 1999) ● Provide students with immediate input and feedback
Automatic speech recognition (ASR) and pronunciation program	A technology that allows a computer to identify the words a person speaks into a microphone. ASR is often a component of speech pronunciation software, and as such, identifies particular parameters of the learner's output, such as prosody or specific sounds,	● Compare student's pronunciation acoustically with a target pronunciation and provide feedback ● Provide learner with an opportunity to work on speaking ability individually, at self-selected pace

(continued)

Table 1. (Continued).

		Example affordances for language study
	and provides feedback on these aspects of performance	● Allow learner to practice simulated dialogue with computerized agent
Network-based social computing		
Virtual world or serious game	A virtual world is a program that allows learners to move a representation of a character, or “avatar”, through a 3-D graphical environment. A serious game is a virtual environment or traditional computer game in which activities are guided or restricted by the program and users have a specified goal or set of goals to complete	● Provide virtual meeting spaces ● Enable learner to navigate within simulated environments, including those modeled after target language locales and incorporating culturally relevant objects ● Encourage role play through the ability to embody different characters within a scenario
Chat	A form of synchronous computer-mediated communication; either text-based or include audio	● Record logs of interactions, which can be printed for review and used as an assessment tool (Tudini, 2003) ● Enable communication and collaboration among students or between students and native speakers without constraints of distance or location
Social networking	Social networking, of which Facebook and MySpace are the best-known examples, enables peer-to-peer communication and collaboration. Users develop their own presence on social networking by creating profile pages about themselves, and then joining networks based on geography, interests, associations, or friendships	● Support networking among users as well as the ability to communicate with others with similar interests ● Enable interaction with native speakers and other students of the target language ● Allow for synchronous and asynchronous communication
Blog	A web application that displays entries authored by the blog owner with time and date stamps and is visible to other web users	● Support personal journaling or blogging and enable feedback in the form of comments on blog posts ● Encourage collaborative learning

(continued)

Table 1. (Continued).

		Example affordances for language study
Internet forum or message board	An asynchronous system in which messages are sent to multiple recipients. Messages are threaded according to topic and a notification is often sent to a user's e-mail address when an update is posted	• Organize discussions via topic thread • Enable online information exchange without constraints of time and distance
Wiki	A website that allows multiple users to post or edit information	• Help students and instructors to find information easily through organization by topic • Enable collaboration on in-class projects • Support open-editing of content
Mobile and portable devices		
Tablet PC or PDA	A tablet personal computer (tablet PC) is a portable personal computer with a touchscreen. A personal digital assistant (PDA) is a hand-held mobile computing device that combines many features now common to other mobile devices: a calendar, contacts list, word processing, and depending on the OS, applications, such as Excel, PowerPoint, and Adobe Reader	• Enable handwritten computer input in target language scripts • Capture diagrams and illustrations digitally • Run software for language learning on portable device
iPod	A portable media player produced by Apple, Inc. It can also serve as external data storage devices with a wide range of memory capacities. iPods can be used to play downloaded television shows and movies and have a small screen for viewing this media. Podcasts, or audio and video digital-media files, can also be downloaded for use with the iPod or other digital media player	• Serve as a portable hard drive for data upload and download • Enable rich input through language-learning podcasts and broadcasts of authentic speech • Enable students to record speech samples or homework activities digitally, and upload these for teacher or peer review
Cell phone or smartphone	A cell phone is a mobile telephone and a smartphone	• Provide mobile internet access

(continued)

Table 1. (Continued).

	Example affordances for language study
is a mobile phone with advanced capabilities, and often, PC-like functionality. A smartphone often has a keyboard or other text entry functionality, internet and e-mail abilities, and the capacity to run an operating system and related software	• Support language-learning activities that utilize text messaging, and sharing and taking pictures • Facilitate teacher–student and student–student communication during remote learning activities

Strength of evidence

In this review, we make a distinction between strong, moderate, and weak empirical support for generalizable claims regarding the effectiveness of technology use in FL learning and teaching. Table 2 presents the three levels of support and the definitions of each level. “Non-experimental” includes observational, case studies, and those studies that included a pre–post-treatment design, but did not include a control group. “Experimental” includes studies with treatment and control groups with subjects randomly assigned to these groups.

The evidence

Schoolhouse or classroom-based technologies: curriculum-driven tools for instruction and assessment

Course/learning management system (CMS)

To our knowledge, only one published study has compared the effectiveness of CMS use and traditional, non-technological alternatives. Sanprasert (2009) examined the effect of CMS use on language learner autonomy in a study of 57 Thai English as a Foreign Language (EFL) learners who were split into experimental (CMS use) and control groups (no CMS, but same learning materials). Based on the quantitative analyses of pre-/post-course questionnaires and the qualitative analysis of learner journals, Sanprasert concluded that CMS users became more independent and more confident learners, and that CMS use can help to develop a learner’s sense of autonomy.

Despite the dearth of evidence on CMS efficacy, S. Carey (1999), in a case study of WebCT use for a graduate-level FL seminar, described CMS as “the most significant innovation that I have experienced to improve the quality and depth of student involvement in a graduate seminar while promoting the students’ improvement in academic reading comprehension and academic writing production” (p. 379). However, as Kvavik (2005) cautioned, “[t]he interactive features [of CMS] least used by faculty were the features that students indicated contributed the most to their learning,” including “... sharing materials with students, ... faculty feedback on assignments, ... and online readings” (p. 7.14).

Table 2. Strength of empirical support for claims of effectiveness.

Strength	Definition
Strong	• Three or more corroboratory well-designed experimental, quantitative non-experimental, qualitative, or mixed methods studies
Moderate	• A single well-designed experimental, quantitative non-experimental, qualitative, or mixed methods study; OR • Two or more well-designed experimental, quantitative non-experimental, qualitative, or mixed methods studies with partially contradictory evidence; OR • Two or more experimental, quantitative non-experimental, qualitative, or mixed-methods studies with design limitations, such as a low number of participants
Weak	• A single well-designed study of any kind with contradictory evidence or with design limitations; OR • Expert opinions based on theory or own practice but not empirical data; OR • Studies with flaws in methodology or methodology not discussed in detail

Interactive white board (IWB)

In the field of FL learning and teaching, the majority of evidence on the effectiveness of IWB use comes from qualitative studies that analyzed self-report data. In some cases, self-report methods, such as questionnaires or interviews, were paired with classroom observations. Although no studies have reported learning outcomes, there is some evidence regarding the impact of IWBs on the process of learning and on learner affect. For instance, use of an IWB improved students' reported ability to memorize material, promoted independence in learning, and encouraged more practice and recycling of already learned material. In addition, instructors teaching languages with non-Latin scripts reported that the IWB was an effective tool to teach the four skills, increase time on task, and encourage active learning.

Both teachers and students were overwhelmingly positive about the IWB and believed that it enhanced learning and teaching. Language teachers felt that the use of the IWB had a positive and sometimes dramatic effect on their teaching and changed their roles in the classroom (Gray, Hagger-Vaughan, Pilkington, & Tomkins, 2005), that the IWB brought a "wow" factor to the classroom (Orr, 2008), and that it increased students' enthusiasm, interest, and engagement in the learning process (Tozcu, 2008), and attracted attention (Schmid, 2007).

ePortfolio

Existing evidence for the effectiveness of ePortfolios for FL learning comes from qualitative studies and is based on participant self-report; therefore, no learning outcomes were measured in these studies. In general, the few studies that exist show a mixture of advantages and disadvantages of the technology. For example, although teachers piloting the European Language Portfolio reported that maintaining ePortfolios is extremely time-consuming, that there are still some implementation barriers, and that students lack motivation and willingness to use ePortfolios (Little & Perclova, 2001), learners in the same study expressed positive reactions towards self-assessment and said that ePortfolios helped them to reflect on their language abilities and knowledge. Additional mixed reactions from pre-service EFL teachers were collected by Kocoglu (2008), whose participants valued

the collaboration enhanced by ePortfolios and the fact that with ePortfolios, they could more easily keep current in their profession. They did not, however, believe that ePortfolios are important in developing reflective thinking. Very positive reactions about the use of ePortfolios by eighth grade students in their EFL class were collected by Chang, Wu, and Ku (2004) and positive assessment results by different stakeholders (students, teachers, developers) were presented by Cheng (2008–2009).

Individual study tools: self-contained programmed applications

Corpus

Claims about corpora and how they can affect language learning, especially those coming from corpus linguists, are stronger than actual evidence for their efficacy. Available evidence regarding the effectiveness of corpus-based materials in FL learning and teaching comes from qualitative studies that use self-report data, or from case studies. Some learners enjoyed using corpora and thought that the technology was beneficial (Farr, 2008) and believed that the use of corpora enhanced their language awareness, improved their command of lexicogrammatical rules and patterns, increased their awareness of context, and promoted discovery learning (Liu & Jiang, 2009). However, learners also expressed their concerns about technological skills needed to use the software and the amount of time needed to adequately use corpora (Farr, 2008). In addition, students had difficulty with corpus-based lexicogrammar learning; in particular, effectively analyzing concordance data (Liu & Jiang, 2009). Some self-report evidence also suggests that the process of learning is changed as a result of corpora consultation, e.g. students tend to check their writing more often while composing (Yoon, 2008).

Some studies suggested that although corpora may be successfully used by linguistics students or students at the advanced level, they may be problematic for students at lower levels (Kennedy & Miceli, 2001). However, some limited evidence is available that beginning students are able to formulate basic grammar rules based on corpus consultation (St. John, 2001).

Electronic dictionary

Numerous quantitative and qualitative studies have been conducted to measure the effectiveness of electronic dictionaries (as opposed to traditional paper dictionaries) for FL reading. These studies have examined how the use of electronic dictionaries affects the efficiency or rate of reading, reading comprehension, retention of incidentally learned vocabulary, or learner attitudes toward reading in a FL. In general, learners who used electronic dictionaries were significantly faster at completing reading tasks than were users of paper dictionaries (Aust, Kelley, & Roby, 1993; Koyama & Takeuchi, 2007; Leffa, 1993). However, in contrast, an earlier study by Koyama and Takeuchi (2004) found no significant difference in time needed to complete work when comparing users of the two types of dictionaries.

The users of electronic dictionaries looked up more words than the users of comparable paper dictionaries (Aust et al., 1993; Koyama & Takeuchi, 2004), but this increased rate of look-up did not disrupt the reading process itself, with students from both types of dictionaries showing similar reading comprehension scores (Aust et al., 1993; Koyama & Takeuchi, 2007; Laufer & Levitsky-Aviad, 2006). In fact,

Leffa (1993) found that learners using electronic dictionaries understood significantly more (86.10 vs. 62.70%) than the users of paper dictionaries and that the use of electronic dictionaries by beginning language students helped to close the gap between stronger and weaker learners. This latter finding was corroborated by Knight (1994), who found that weaker learners benefited more in terms of comprehension and vocabulary learning than did stronger learners, although her study only investigated electronic dictionary use by learners with different aptitudes and did not contrast this with paper dictionary use.

Several studies also examined the rate of retention of words looked up in electronic dictionaries and reported mixed results. For instance, Laufer and Hill (2000) found no evidence that more look-ups increased retention of looked-up words and Koyama and Takeuchi (2004) found significantly better retention for the users of paper dictionaries. Peters (2007), who only investigated electronic dictionary use, showed a strong correlation between task-relevance and retention of looked-up words, and argued that her results do not support claims that the ease of electronic dictionary use induces shallow processing.

Several studies examined the effect of electronic dictionaries on learner attitudes toward reading in a FL and found that learners strongly prefer using electronic dictionaries over paper dictionaries, and have a more positive attitude and willingness to read in a FL when using electronic dictionaries (Aust et al., 1993; Laufer & Levitsky-Aviad, 2006; Liou, 2000; Loucky, 2005). In contrast, Koyama and Takeuchi (2004) found no preference for electronic dictionaries over paper dictionaries among their subjects.

Electronic glosses and annotations

An advantage of using electronic glosses rather than using electronic dictionaries is the possibility of providing context-specific definitions or translations. Participants in Chun's (2001) study, which compared the use of electronic dictionaries with that of tailored glosses, stated that glosses did not require them to recreate the dictionary form of the word in order to look it up, a capability not possessed by electronic dictionaries. In the same study, researchers noted that the electronic dictionary users also had to decide among multiple definitions or translations for a word.

Two meta-analyses compared the effect sizes obtained in computer-based gloss studies with those from paper-based gloss studies and found larger effect sizes on L2 reading comprehension for the studies involving electronic glosses (Taylor, 2006, 2009). Hong (1997) directly compared the use of computer-assisted reading with the conventional paper-dictionary method, and found that learners using a multimedia program that gave them access to sound files, Chinese glosses, and English definitions completed the tasks in half the time and had higher comprehension scores on multiple-choice tests.

Most quantitative and qualitative studies of the effectiveness of electronic glossing or annotations for FL reading have compared the effects of different types of electronic glossing on incidental vocabulary learning or reading comprehension. In general, a combination of text + picture glosses seems to be more effective than text-only glosses (Chun & Plass, 1996; Kost, Foss, & Lenzini, 1999; Plass, Chun, Mayer, & Leutner, 1998; Yeh & Wang, 2003; Yoshii, 2006; Yoshii & Flaitz, 2002) or picture-only glosses (Kost et al., 1999; Plass et al., 1998; Yoshii & Flaitz, 2002). Results for more complex types of annotations are mixed. For example, whereas Al-

Seghayer (2001) found that text + video annotations were more effective than text + picture annotations, Chun and Plass (1996) found the opposite, and Yeh and Wang (2003) found that text + picture glosses were more effective than glosses that included text + picture + sound. In addition, some studies revealed that learners did not take advantage of multimedia glosses and annotations (Davis & Lyman-Hager, 1997; Lomicka, 1998) and that annotations were not helpful (Ariew & Ercetin, 2004).

Studies that compared the use of L1 dictionary-form glosses and sentence-level translation glosses found no difference in text comprehension, but the dictionary-gloss group took more time to read the text (Gettys, Imhof, & Kautz, 2001). However, Grace (2000) found that the availability of L1 dialogue-level translation glosses significantly improved performance on both immediate and delayed vocabulary posttest scores as compared to L2 explanation glosses. When the effectiveness of single L1 translation glosses was compared with interactive glosses (a choice between two L1 translations accompanied by immediate feedback), the results showed that the interactive glosses were significantly more effective for learning both vocabulary and grammatical structures as measured by the immediate posttests, but that only grammatical structures maintained a significant difference in performance on the delayed posttest (Nagata, 1999).

Three studies reviewed also included qualitative data on learners' attitudes toward computerized glosses and annotations. Students found annotations useful; felt that the hypermedia reading environment was enjoyable, interesting, and easy to understand (Ariew & Ercetin, 2004); reported favorable impressions of computerized glosses (Lenders, 2008); felt that glosses made them more independent (Davis & Lyman-Hager, 1997); and felt that glossed words were helpful and led to faster reading (Chun, 2001).

Intelligent tutoring system (ITS)

Some evidence exists for the effectiveness of ITS in language learning in relation to intelligent feedback.² Nagata (1993) performed a study, results of which led her to conclude that intelligent computer feedback is more effective than traditional feedback, especially in the case of particle acquisition. In another study, Nagata (1997) compared intelligent, rule-based deductive feedback generated by NLP with example-based inductive feedback and concluded that the intelligent feedback is more effective for learning complex structures.

Dodigovic (2007) performed an evaluation of an ITS designed to address seven errors common to learners of English as an L2 and found that use of an ITS resulted in an average 83% reduction in these types of errors. While extraordinarily positive at first glance, interpretation of these results is complicated by a number of confounds, including the use of different formats for pretest and posttest and the lack of a control group. In spite of these limitations, the study highlights a number of ITS features that are both grounded in SLA theory and seem to function effectively. For example, the ITS targets error-types identified as common and problematic through analysis of genuine texts produced by learners; it implements instruction within a communicatively focused exercise, and it provides immediate corrective feedback to learners. Similarly, MacWhinney (1995) cited communicative context, corrective feedback, and the diagnostic value of learner errors among his recommendations for bringing ITS design more into line with the lessons learned

from experimental psychology, psycholinguistics, and SLA research in recent decades.

Another application of ITS has been with automatic speech recognition (ASR). The software *CandleTalk* was studied with 49 college-level ESL learners in Taiwan. The mean total score for oral performance improved significantly across all participants between pretest and posttest, but intelligibility did not improve (Chiu, Liou, & Yeh, 2007). Students also significantly improved reading and speaking scores in Arabic assessed through pretest and posttest face-to-face interviews with Defense Language Institute (DLI) raters and mock Defense Language Proficiency Tests (DLPT) using the Interactive Drama Inc. Virtual Conversations software (Harless, Zier, & Duncan, 1999). Students noted an increase in motivation while using the software and improved speaking confidence as a result of participating in this training. Pilot testing of the Military Language Tutor for Arabic also produced positive results with significant pretest–posttest learning gains in vocabulary, grammar, pronunciation, and fluency using *Dragon* speech recognition software (Holland, Kaplan, & Sabol, 1999). While a common complaint in all of these studies is that the ASR accuracy is not 100%, students typically reported positive experiences when using the software, increased motivation to practice the language and an increased confidence in their ability to use the language. These affective variables and the promising empirical results suggest that ASR can be a valuable component of ITS for FL learning.

Grammar checker

The literature on electronic grammar checkers suggests that language learners need training in order to use this technology effectively. Jacobs and Rogers (1999) compared the use of Microsoft Word's built-in French grammar checker with the use of paper grammar references. Prior to an explicit training session on optimal use of the grammar checker, the group using the paper references outperformed the group using the grammar checker. After the training session, however, the two groups performed with comparable accuracy.

Burston (2001) reached a similar conclusion regarding the French grammar checker Antidote, pointing out one particularly helpful strategy: when the grammar checker remarks that it cannot parse a full sentence, the learner must manually highlight the sentence's fragments and run the checker over each one individually. Learners, who had been trained in the use of the technology, including this manual parsing strategy, outperformed a group who used no grammar checkers across three writing assignments, when compared on morphosyntactic accuracy.

Automatic speech recognition and computer-assisted pronunciation training (CAPT)

In the field of language learning, ASR is used as a part of CAPT software to improve learners' FL pronunciation. So far, studies of these technologies have produced mixed results. Promising results have been found with programs that record student speech and acoustically analyze it, comparing the student's pronunciation and prosody to a native speaker sample using visual feedback. Although students often require additional training so that they can interpret the feedback, such programs can improve students' prosody and vowel pronunciation (M. Carey, 2004; Hardison, 2004). Kay Elemetrics' *Computer Speech Lab* has been used with college-level students of both French and Japanese to improve learners' prosody, pitch, and

duration (Hardison, 2004; Hirata, 2004). Significant improvements on pretest and posttest measures of prosody and segmental accuracy were found in the experimental groups with generalization to novel sentences. Students improved in both production and perception of these distinctive features, while the control group did not experience the same gains (Hirata, 2004). Students reported an increase in their speaking confidence and an increased awareness of prosody (Hardison, 2004). Another *Kay* product, *Sona-Match* was used to look at the effect of visual feedback on the pronunciation of vowels by ESL learners. Students in the experimental group experienced a significant improvement in their pronunciation after 5 h of instruction with visual feedback (M. Carey, 2004). An oral reading technique, Cued Pronunciation Reading was applied in a study with 75 ESL learners, which revealed that participants from the treatment group made significant gains in the perception of pausing, perception of word stress, and controlled production of word stress (Tanner & Landon, 2009).

Several studies have looked at the accuracy of the feedback for pronunciation provided by the CAPT software as compared to human judgment of pronunciation accuracy and found fairly low correlations between the software scores and the human scores (Kim, 2006; Machovikov, Stolyarov, Chernov, Sinclair, & Machovikova, 2002; Rypa & Price, 1999). This correlation is also consistent with the level found when comparing scores between human judges (Rypa & Price, 1999). The software appeared even less reliable when comparing intonation scores given by the software with judges' ratings of intonation, in which case the correlation coefficient was nearly zero, indicating that ASR and CAPT still have a long way to go before it can replace human interaction (Kim, 2006).

Network-based social computing technologies: synchronous computer-mediated communication (CMC)

Virtual world/serious game

There is no clear evidence that learning in virtual worlds is more effective than traditional forms of classroom learning or other forms of distance learning. Existing studies have primarily reported affective reactions to the use of virtual worlds or students' opinions of the technology's utility. For example, students may report that some aspect of their language learning was improved through use of the technology (O'Brien & Levy, 2008; Sykes, 2008), or may indicate that they enjoyed learning in a virtual environment (Shih & Yang, 2008). Although some of these studies provide evidence that students were able to improve particular aspects of targeted language skills through lessons in a virtual world environment (Renalli, 2008; Sykes, 2008), the lack of control groups using traditional methods make it impossible to evaluate whether learning was *improved*, *impaired*, or *unaffected* by these technologies.

In terms of serious games, the most relevant literature addresses the use of immersive games for military training, such as *America's Army* or the Navy's *Strike and Retrieve* (de Freitas, 2006). For language learning, the *Tactical Language and Culture* systems available from Alelo, Inc. (<http://www.tacticallanguage.com/>) are the most direct application of this technology. Surface, Dierdorff, and Watson (2007) evaluated the effectiveness of the Tactical IraqiTM program with military personnel. Analyses of pre- and post-program multiple-choice testing indicated that learners demonstrated significant improvement in their declarative knowledge of Iraqi Arabic

language and culture after 40 h of Tactical Iraqi use (plus classroom training for some subjects) (pp. 26–27).³ Due to impending deployments, no users completed the entire recommended 80-h program of instruction. These results suggest that programs such as Tactical Iraqi™ can be useful for increasing soldiers' knowledge of language and culture, but the authors recommend that they be part of a "structured language program," rather than as a stand-alone training solution. Interestingly, however, students rated the traditional computer-training element of the program as more useful and more enjoyable than the 3-D videogame element. Students felt that the program was most useful for learning vocabulary, practicing the language, and receiving feedback while instructor-led lessons were ranked higher in terms of learning and practicing cultural knowledge, overall training, and mission-related training.

One additional study, which used a more traditional video game for English learning, indicates that video game players learned less target vocabulary than did students who sat nearby and passively watched the same game, perhaps due to the cognitive load required to work the game (deHaan, Reed, & Kuwada, 2010).

Chat

A few studies have specifically compared the efficacy of text or voice chat to more traditional forms of classroom discussion or to other control groups. Sykes (2005) found that Spanish learners who used written chat to practice pragmatic skills produced more complex output and used a wider variety of pragmatic strategies at posttest than did students in either voice chat or face-to-face discussion practice groups. Sykes hypothesized that this effect may be due to learners' having more time to construct, and so practice, complex structures because of the natural delay inherent in written chat interactions. Several studies found similar results for the development of oral proficiency, with students who practiced with written chat improving more than groups who practiced with face-to-face discussion (Blake, 2009; Payne & Ross, 2005; Payne & Whitney, 2002). In addition, Payne and colleagues (Payne & Ross, 2005; Payne & Whitney, 2002) provide preliminary evidence that written chat may be used as a compensatory mechanism by learners with relatively low levels of phonological working memory. Abrams (2003) also examined the usefulness of written chat for the development of oral proficiency, comparing a group of students who used chat to two other groups: one which used asynchronous CMC for discussions (bulletin board) and one which used no CMC. Abrams found that students who practiced with written chat subsequently produced the greatest quantity of output in posttest face-to-face discussions, whereas those who used asynchronous chat produced the least amount of output. Other measures of posttest oral proficiency, however (lexical density and syntactic complexity), did not vary significantly between groups. Similarly, Satar and Özdener (2008) found that both voice chat and written chat practice groups outperformed control participants in a posttest measure of speaking ability and the written chat group also had a drop in FL anxiety.

Additional research suggests that many FL students are more likely to communicate through chat than in the traditional large group, face-to-face classroom, (Kern, 1995; Sullivan & Pratt, 1996) and that students using chat produce more sentences and more total words, and use a greater number of

complex sentences and complex morphosyntactic structures (Kern, 1995; Warschauer, 1996).

Because of its written format, text-based chat may promote “noticing” – an important theoretical construct in SLA research that is thought to be necessary for successful learner intake in the L2 (Chen, 2008; Kitade, 2000; Shekary & Tahririan, 2006). Written chat may also increase students’ focus on form, and increase the salience of student errors and interlocutor feedback (Lai & Zhao, 2006; Lee, 2008; Shekary & Tahririan, 2006); but see Loewen and Reissner (2009) for contradictory results finding more focus-on-form in a face-to-face discussion context.

Network-based social computing technologies: Web 2.0 tools and asynchronous communication

Asynchronous CMC allows students to communicate and collaborate via the computer without the constraints of time and distance. This affordance has made asynchronous communication valuable to FL teachers and students, and has been noted in many studies looking at blogs, wikis, Internet forums, and discussion or message boards. Although asynchronous communication reduces opportunities for negotiation of meaning, it affords greater time for processing of language input, thoughtful TL output, and learner self-correction (Kitade, 2008; Levy & Stockwell, 2006; Warschauer, 1997). As described in the following sections on individual technologies, in general, the available empirical evidence supports the viability of asynchronous communication as a mode of FL learning, rather than providing solid evidence of its effectiveness.

Social networking

The effects of social networking on language learning and the implications for instruction and curriculum development are as yet unknown. In reviewing the literature, we did not find any studies on the use of social networking for language learning that included data on language use (e.g. number of contributions per user, length of contribution, types and distributions of negotiations and feedback). Debski (2002) strikes a sobering note on the lack of studies concerning social computing (including social networking) for language learning:

... social computing ... has probably been the single most important factor changing L2 learning and teaching practices in recent time. Second language students are asked to communicate and collaborate in the target language with overseas partners, to search for information on the Internet, create Web projects and share them with online communities ... Despite the growing popularity and intuitive appeal, the position of computer-supported collaborative learning is however still far from settled and exactly what and how students learn through such practices is still unclear. (p. 130)

Blog

As blogs are relatively new in the FL classroom, very few studies have quantitatively examined how they can improve FL productive and receptive skills in comparison to traditional writing assignments. Student writing has been found

to improve when blogs were incorporated into the curriculum; however, no studies included a control group. Changes in L2 writing included the use of new phrases, improvements in spelling and the use of accent marks, and an increase in the use of conjunctions (Thorne, Webber, & Bensinger, 2005, as cited in Thorne & Payne, 2005). Other noted changes in L2 writing included improved accuracy of verbal morphology and more extensive production (Armstrong & Retterer, 2008).

In qualitative studies, students have indicated that they prefer blogging to traditional journals or weekly essays, that they find publishing their writing online to be motivating, and that they believe that blogging improves their writing (Armstrong & Retterer, 2008; Thorne et al., 2005). In other studies, students commented that reading and commenting on blogs written by native speakers increased their confidence in writing the TL, provided opportunities to reflect on their own L2 production and proficiency gains, and elicited greater creativity than in traditional writing assignments (Ducate & Lomicka, 2005, p. 419, 2008; Thorne & Payne, 2005). Students were also observed to experiment with the TL in ways not observed in traditional writing assignments (Ducate & Lomicka, 2008). These students saw blogging as an enjoyable part of their language curriculum but noted that they would have preferred more flexibility in their choice of blog topic, and suggested that creativity be included in the grading rubric to encourage maximal use of blog capabilities (Ducate & Lomicka, 2008).

In a rare case study of voice-blogging, Sun (2009, p. 99) concluded that "...students perceived blogging not only as a means of learning, but also as a means of self-presentation, information exchange, and social networking" and that "blogs constitute a dynamic forum that fosters extensive practice, learning motivation, authorship, and development of learning strategies." However, as Comas-Quinn and Mardomingo (2009) noted in a descriptive study of mobile blogs in language learning, "Assessing and evaluating the learning that results from this kind of activity is a real challenge, as the creativity and subjectivity involved will result in outcomes which differ enormously from learner to learner, and which are difficult to measure" (p. 109).

Internet forum and discussion/message boards

Asynchronous communication using forums and discussion/message boards, similarly to synchronous chat, enables users to communicate in the TL or collaborate on a project related to the TL. Message boards also afford opportunities to provide feedback on learner output. In a qualitative study, students reported that they valued the feedback they received from their native speaker partner on language form. However, this feedback was only provided in the e-tutoring sessions when it was explicitly required and not during project collaboration (Ware & O'Dowd, 2008). This finding has pedagogical implications for instructors who wish to use bulletin boards for peer communication, and implies that instructors will need to design the task so feedback is considered an integral part of the interaction by the students.

Wiki

Limited empirical evidence exists for the use of wikis in FL education, mostly in the form of case studies that explore the use of wikis for collaborative L2 writing

processes. Some case studies have noted an uneven distribution of participation, with a small number of students actively contributing in L2 collaborative writing, and with greater attention paid to adding, deleting, clarifying, or elaborating information, and changing fonts and colors. Less attention was directed to writing acts that require more critical thinking, such as analyzing grammatical accuracy and synthesizing information (Kessler, 2009; Kessler & Bikowski, 2010). Not surprisingly, teacher investigators have concluded that teachers must be actively involved in student contributions to the wiki to ensure collaboration between students is successful (Ioannou & Artino, 2008; Lund & Smørdal, 2006).

However, other case studies have provided evidence that L2 learners using wikis collectively address TL errors and offer solutions (Lee, 2010). Additionally, writing for a wider audience than just the instructor made students more interested in what they produced the process of editing and revising particularly helped them write better in-class individual compositions at the end of the course (ibid). L2 wiki writing may be particularly beneficial for collaborative L2 to L1 translation work, with learners showing progress in their ability to differentiate TL writing styles (Miyazoye & Anderson, 2010). In a note of caution regarding the siren song of new technologies, Lund (2008) observed that it is “the activity and not the technology per se that makes the difference” (p. 50).

Mobile/portable networkable devices for anytime, anywhere learning

Tablet PC and PDA

So far, the effectiveness of tablet PCs for FL learning has not been extensively studied. Our search of the literature uncovered only a single study with any data on the impact of tablets; in this case, on language learning processes. In a study of elementary EFL learners in Taiwan, Lan, Sung, and Chang (2007) conducted a study of the collaborative, peer-assisted learning practices of elementary school EFL learners in Taiwan in small reading groups, with and without tablet PCs. The tablet PC group used a communication software, Skype, and the non-tablet group used no computers. They concluded that compared to learner groups without tablets, the learner groups with the tablets attended more to the reading tasks, and exhibited more collaborative behaviors, such as giving support and feedback, and avoiding conflict. The authors also cite their “distinct impression that [tablet use] seemed to reduce anxiety in elementary EFL learners, promote motivation to learn, and enhance oral reading confidence” (p. 142).

In the case of PDAs, as of tablet PCs, there appear to be two related questions of effectiveness: the question of the viability of the technology to support FL learning, and the question of the effectiveness of the pedagogical approaches implemented through and supported by the technological platform. The research appears to be directed at the former question rather than the latter. Gilgen (2005) and Samuels (2005) report that students considered PDAs useful for reading and writing activities (including written chat) but not for listening or speaking activities. Paredes et al. (2007), who used A Writing eLearning Appliance (AWLA) claim that their application’s “extensive use has validated it as an effective writing tool for learning how to write by writing” (p. 955) but offer no evidence of language learning effectiveness.

In a small-scale (15 learners) study of PDA use with a personalized intelligent mobile learning system with a learning outcome measure, Chen and Hsu (2008)

found significant gains in TL reading comprehension. Using the same participants and research design, Chen and Chung (2008) evaluated a PDA-based vocabulary learning system designed to tailor English vocabulary learning materials to individual learners, and found that TL vocabulary recall on a cloze exercise improved significantly compared to a pre-study test. However, neither of these studies included a control group without PDAs. Liu (2009) conducted a case study with 64 Chinese EFL high-school learners, using a context-aware ubiquitous learning environment for language listening and speaking. Students were randomly assigned to an experimental group using PDA phones that used radio frequency identification to find items in the students' location to make relevant English language learning activities and materials available. The control group used equivalent printed materials and CD players. Analyses of pretest and immediate posttest scores showed a large, significant improvement in listening and speaking skills for users of the context-aware ubiquitous learning activities, compared to students using similar content on CDs and printed materials. Liu concluded that the positive results showed that PDAs can be useful for mobile language learning.

iPod

Although we found no empirical evidence of the effectiveness of iPod use in the language classroom, there is some data on students' affective or motivational responses to the use of the devices. For example, from a set of 121 language students who used iPods with microphones in the language classroom, 90% of students reported that they liked the convenience of the iPods, and 91% of students said that "overall" they benefitted from working with an iPod. In addition, 80% of students reported no trouble with technological glitches. Fifty-one percent thought that "access to the iPod increased knowledge of TL," 67% reported that "having the iPod motivated me to spend more time" on listening/speaking activities for language class, and 56.7% thought that "using the iPod helped me learn the language better." Despite these indicators that students approved of iPod use, no correlation was found between students' reported attitudes toward the technology and their classroom grade point average (GPA) (Sathe & Waltje, 2008). Abdous, Camarena, and Facer (2009) looked at whether using podcasting in the language classroom in an integrated (as opposed to a supplemental) manner would lead to greater efficacy and frequency of use, as assessed by students' ratings. The authors defined "integrated use" in terms of the breadth of class activities into which podcasting was incorporated, as well as the degree of planning involved. In the 113 anonymous self-reports they collected, students in the integrated podcasting condition provided higher ratings regarding how frequently they used the podcasts, their facility, and effectiveness.

Cell phone/smartphone

Empirical studies of the use of mobile phones in language instruction have focused primarily on the use of the Short Message System (SMS) feature. In all cases, students in the SMS group significantly outperformed students from Web and paper groups on immediate posttest vocabulary recall measures (Lu, 2008; Thornton & Houser, 2002, 2003, 2005). However, in Lu's study, a delayed posttest no longer

detected any difference between the SMS and paper conditions in terms of vocabulary recall.

Another group of studies compared students' accuracy and speed in completing language learning activities with some students using SMS and others a PC. These studies revealed that vocabulary drills completed via SMS consistently took more time (Stockwell, 2009b). Although this may have been a result of limitations imposed by the compact format of the mobile devices, slower performance may also have resulted from students completing the vocabulary drills in the midst of otherwise distracting environments. Similar results were found by Kiernan and Aizawa (2004) who reported that, when comparing PC e-mail users, mobile phone e-mail users, and face-to-face speakers, both e-mail groups were slower than those students using face-to-face communication. These differences were attributed to the relative speed of typing vs. speaking, and the relative speed of typing on mobile thumb pads vs. keyboards. In general, fewer words were used by mobile phone e-mail users, yet they were able to communicate effectively.

Students' attitudes were also measured. The vast majority preferred the SMS instruction, wished to continue such lessons, and believed it to be a valuable teaching method (Thornton & Houser, 2002, 2003, 2005).

Evidence summary

In spite of the abundance of publications available on the topic of technology use in FL learning and teaching, the evidence that the technology has made a measurable impact upon FL learning or teaching is quite limited. Existing studies span a wide range in terms of validity and reliability and many do not include measures of outcome data.

In our search for evidence of effectiveness of technology for FL learning and teaching, we focused on studies that compare the use of technology with more traditional non-technological methods or materials. For clarity of presentation, we categorized the reviewed studies into those concerning enhanced input and comprehension, output and interaction, feedback, affect and motivation, metacognition, and metalinguistic knowledge (see Table 3). Among those technologies that were included in our review, the only strong support⁴ we found for an impact of technology on FL learning and teaching were for the ASR programs and chat. Research shows that the ASR technology can facilitate improvement in pronunciation to a larger extent than human teachers can and, because of constant improvements of this technology, ASR programs have great potential in FL learning. More strong evidence came from a number of studies investigating the use of chat in FL learning. These studies proved that with chat, both the amount of learners' language production and its complexity significantly increased.

The literature revealed moderate support for the claim that technology use changed the process of learning; for example, it caused more frequent dictionary look-ups or faster completion of tasks. However, further investigations determined that increased frequency of look-ups did not make a significant difference in learning outcomes. The literature also revealed moderate support for a number of other claims, such as demonstrating pretest–posttest gains in speaking, reading comprehension, vocabulary, grammar, and fluency; enhancing L2 speaking proficiency; acquiring more vocabulary words than with more traditional methods; providing effective feedback to learners; and promoting noticing and focus on form.

Table 3. Levels of support for research claims by area and technology type.

Claim	Supporting study or studies	Technology type	Support
Enhanced input and comprehension			
Technology enhances learners' comprehensibility of input	Lefia (1993)	Electronic dictionary	Weak
Technology enhances L2 reading comprehension	Taylor (2006, 2009) Chen and Hsu (2008)	Electronic gloss and annotation PDA	Weak Weak
Enhanced output and interaction			
With technology, learners improve pronunciation efficiently	M. Carey (2004); Hardison (2004); Hirata (2004)	Automatic speech recognition	Strong
With technology, learners' language production increases, both in terms of amount and complexity	Kern (1995); Sullivan and Pratt (1996); Warschauer (1996)	Chat	Strong
With technology, learners demonstrate pretest–posttest gains in different areas, including speaking, reading comprehension, vocabulary, grammar, fluency	Chiu et al. (2007); Harless et al. (1999); Holland et al. (1999)	Intelligent tutor	Moderate
With technology, learners complete tasks faster than without technology	Aust et al. (1993); Koyama and Takeuchi (2007); Lefia (1993)	Electronic dictionary	Moderate
With technology, frequency of dictionary look-ups increases	Hong (1997)	Electronic gloss and annotation	Weak
Technology enhances L2 speaking proficiency	Aust et al. (1993); Koyama and Takeuchi (2007) Blake (2009); Payne and Ross (2005); Payne and Whitney (2002); Satar and Özden (2008)	Electronic dictionary Chat	Moderate Moderate
With technology, learners acquired more vocabulary words than with Web- or paper-based training	Thornton and Houser (2002, 2003, 2005)	Cell phone (SMS)	Moderate
Web- or paper-based training			
With technology, accuracy of learners' FL production improves	Dodigovic (2007) Burston (2001); Jacobs and Rogers (1999)	Intelligent tutor Grammar checker	Weak Weak

(continued)

Table 3. (Continued).

Claim	Supporting study or studies	Technology type	Support
With technology, learners attend more to reading tasks	Lan et al. (2007)	Tablet PC	Weak
	Ioannou and Artino (2008); Lund and Smørdal (2006)	Wiki	Weak
Technology facilitates collaboration	Lan et al. (2007)	Tablet PC	Weak
	Yoon (2008)	Corpus	Weak
With technology, frequency of edits of own writing increases	Ducate and Lomicka (2008)	Blog	Weak
With technology, learners tend to experiment with TL in ways not observed in traditional writing assignments			
With technology, learners can increase their knowledge of TL language and culture	Surface et al. (2007)	Serious game	Weak
Enhanced feedback			
Intelligent feedback is more effective than traditional feedback	Nagata (1993, 1997)	Intelligent tutor	Moderate
With technology, learners can obtain immediate corrective and targeted feedback	Dodigovic (2007)	Intelligent tutor	Weak
Enhanced affect and motivation			
Learners prefer working with technology over traditional non-technological materials	Armstrong and Retterer (2008)	Blog	Moderate
	Aust et al. (1993); Laufer and Levitsky-Aviad (2006); Liou (2000); Loucky (2005)	Electronic dictionary	Moderate
With technology, learners are more motivated and engaged in the process of learning	Gray et al. (2005); Orr (2008); Tozeu (2008)	Interactive white board	Weak

(continued)

Table 3. (Continued).

Claim	Supporting study or studies	Technology type	Support
Learners prefer SMS over Web or paper-based methods With technology, students enjoy the process of learning Technology facilitates confidence in producing in TL With technology, students are motivated to spend more time on learning activities	Lu (2008); Thornton and Houser (2002, 2003, 2005)	Cell phone (SMS)	Weak
	Shih and Yang (2008)	Virtual world	Weak
	Davis and Lyman-Hager (1997)	Electronic gloss and annotation	Weak
	Ducate and Lomicka (2008)	Blog	Weak
	Sathe and Waltje (2008)	iPod	Weak
Enhanced metacognition			
With technology, learners are more likely to correct their errors than in face-to-face conversation Technology helps students become independent and confident learners (Asynchronous) technology enables more time for reflection, processing of input, and editing	Lai and Zhao (2006)	Chat	Weak
	Sanprasert (2009)	CMS	Weak
	Kitade (2008)	Bulletin board/asynchronous chat	Weak
Enhanced metalinguistic knowledge			
Technology can promote noticing and focus on form	Chen (2008); Kitade (2000); Lee (2008); Shekary and Tahririan (2006)	Chat	Moderate
Technology enhances learners' command of grammatical rules and language awareness	Liu and Jiang (2009)	Corpus	Weak

A large number of studies confirmed that learners enjoy using technology in FL learning and that they prefer using technology over more traditional methods and materials. Because of technology, learners tend to be more engaged in the process of learning, and have a more positive attitude towards learning. We classify this level of support as moderate, by and large because it is often based on qualitative self-reported and observational data, and because we found that the theme of affect and motivation spread across several technologies, such as IWB, electronic dictionary, blogs, and virtual worlds. Note that in the case of a single study, no matter how strong the results of that specific study, the strength of empirical support listed in Table 3 will, necessarily, be either weak or moderate based upon the definitions of strong, moderate, and weak shown in Table 2. In all cases, the rating under the Support column in Table 3 represents the sum strength of evidence for a particular methodology within the reviewed literature, not a rating of the quality of individual studies. See Appendix 1 for additional information about studies included in Table 3.

Conclusions

Although the use of technology to enhance FL learning and teaching has grown rapidly during the past three decades, most research has focused on its viability for supporting FL learning; very few well-designed empirical studies support its efficacy for improving FL learning processes or outcomes. Rather, most CALL studies seem to focus on either describing the affordances offered by particular types of technology or measuring their effects on students' affective reactions, such as increased motivation or increased enjoyment of learning activities. Although describing technology's uses and students' enjoyment when using it are admirable and useful goals, it remains unclear to what extent the activities supported by the technology or the potential increased motivation attributed to them actually increase students' learning. One might make the argument that, for example, increased motivation and increased options for self-selecting study activities will lead to deeper engagement with language, more time on task, and, thus, increased proficiency. However, for most technologies, actual increases in learning or proficiency have yet to be demonstrated. The domain of CALL has not been systematically investigated, and some potential uses of technology for learning have not been explored at all. Clearly, empirically derived evidence is required to quantify, characterize, and document the impact technology can have on adult FL learning. CALL technologies have now matured to the point where studies of this type are now possible.

Carefully controlled studies of language learning, in contextually rich and naturalistic environments, are not easy to design or to analyze. The challenges to this type of work are very real and we do not underestimate the obstacles that researchers face in this area, including limited sample sizes, convenience samples of existing classes, lack of suitable control groups, confounding of treatment conditions due to pre-existing organizational requirements or scheduling, and the need to measure changes in learning or proficiency over very small increments of time – such as a single semester. Clarity on specific technologies or combinations of technologies and learning activities must, thus, arise from an analysis of converging evidence from multiple studies, including meta-analyses of studies when possible.

This review is an attempt to summarize existing evidence for a variety of technologies, focused especially on those that provide some measure of efficacy. As

such, it provides only a partial view of the larger CALL literature on each technology reviewed. Further work will be required to describe and measure the effects of specific technologies on learners at a variety of proficiency levels, in a variety of learning environments. It is hoped that the current review will provide a useful reference for those wishing to take on this task, as CALL moves away from basic description of what technology *can* do and toward a mature theory of what it *should* do to support teachers and learners, and when and how different technologies can best be used to support learning. Such a theory should relate the affordances of technologies to the learning processes being targeted and the needs of individual students.

However, given that digital networkable technologies have become ubiquitous (or nearly so), the next phase of CALL research should focus on how to optimize L2 learning in two distinct but related areas. The first area involves affordances that technologies offer by applying fundamental principles of learning from SLA and cognitive psychology (for example, optimal retrieval intervals). The second should investigate the specific characteristics of technologies that may have differential effects on long-term storage in memory and retention (for example, visual vs. textual annotations in L2 electronic glosses).

Finally, we advise caution before being led down the golden path of technology. Where technology is used to enhance learning, pedagogical goals and not technological means should come first (Gray, 2008). Using technology in delivering a lesson or instructional unit will not make bad pedagogy good. Nor does a lack of technological tools or applications prevent effective teaching: “Good teaching remains good teaching with or without the technology” (Higgins, Beauchamp, & Miller, 2007, p. 215).

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Notes

1. Promising work is being done in this area; see, for example, Heift and Schulze (2007) or Nagata (2009).
2. Intelligent feedback is defined as a feedback generated with the use of NLP technology (parsing). Traditional feedback does not involve parsing technology and it is generated by character-by-character matching technique.
3. Unfortunately, the authors did not present sufficient details to calculate the effect size of this improvement.
4. Recall the definition of “strong support” in this article is that the efficacy of the technology is supported by “three or more corroboratory well-designed experimental, quantitative non-experimental, qualitative, or mixed methods studies”.

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References

- Abdous, M.h., Camarena, M.M., & Facer, B.R. (2009). MALL technology: Use of academic podcasting in the foreign language classroom. *ReCALL*, 21(1), 76–95.
- Abrams, Z.I. (2003). The effect of synchronous and asynchronous CMC on oral performance in Germ. *Modern Language Journal*, 87, 157–167.
- Al-Seghayer, K. (2001). The effect of multimedia annotation modes on L2 vocabulary acquisition: A comparative study. *Language Learning & Technology*, 5(1), 202–232.
- Ariew, R., & Ercetin, G. (2004). Exploring the potential of hypermedia annotations for second language reading. *Computer Assisted Language Learning*, 17, 237–259.
- Armstrong, K., & Retterer, O. (2008). Blogging as L2 writing: A case study. *AACE Journal*, 16, 233–251.
- Aust, R., Kelley, M., & Roby, W. (1993). The use of hyper-reference and conventional dictionaries. *Educational Technology Research and Development*, 41, 63–73.
- Bernardini, S. (2004). Corpora in the classroom: An overview and some reflections on future developments. In J. Sinclair (Ed.), *How to use corpora in language teaching* (pp. 15–38). Philadelphia: J. Benjamins.
- Blackburn, J.L., & Hakel, M.D. (2006). Enhancing self-regulation and goal orientation with ePortfolios. In A. Jafari & C. Kauman (Eds.), *Handbook of on ePortfolios* (pp. 83–89). Hershey, PA: Idea Group Reference.
- Blake, W.C. (2009). Potential of text-based internet chats for improving oral fluency in a second language. *Modern Language Journal*, 93, 227–240.
- Burston, J. (2001). Computer-based grammar checker and self-monitoring. *CALICO Journal*, 18, 499–515.
- Carey, S. (1999). The use of webCT for a highly interactive virtual graduate seminar. *Computer Assisted Language Learning*, 12, 371–380.
- Carey, M. (2004). CALL visual feedback for the pronunciation of vowels. *CALICO Journal*, 21, 571–601.
- Chang, Y., Wu, C., & Ku, H. (2004). The introduction of electronic portfolios to teach and assess English as a Foreign Language in Taiwan. *TechTrends*, 49(1), 30–35.
- Chappelle, C. (1997). CALL in the year 2000: Still in search of research paradigms? *Language Learning & Technology*, 1, 19–43.
- Chen, W.C. (2008). *Noticing in text-based computer-mediated communication: A study of task-based telecommunication between native and nonnative English speakers* (Unpublished doctoral dissertation). UMI 3333633, Texas A&M.

- Chen, C.M., & Chung, C.J. (2008). Personalized mobile English vocabulary learning system based on item response theory and learning memory cycle. *Computers & Education*, 51, 624–645.
- Chen, C.M., & Hsu, S.H. (2008). Personalized intelligent mobile learning system for supporting effective English learning. *Educational Technology & Society*, 11, 153–180.
- Cheng, G. (2008–2009). Implementation challenges of the English language ePortfolio system from various stakeholder perspectives. *Educational Technology Systems*, 37(1), 97–118.
- Chiu, T.-L., Liou, H.-C., & Yeh, Y. (2007). A study of web-based oral activities enhanced by automatic speech recognition for EFL college learning. *Computer Assisted Language Learning*, 20, 209–233.
- Chun, D. (2001). L2 reading on the web: Strategies for accessing information in hypermedia. *Computer Assisted Language Learning*, 14, 367–403.
- Chun, D., & Plass, J.L. (1996). Effects of multimedia annotations on vocabulary acquisition. *The Modern Language Journal*, 80, 183–198.
- Comas-Quinn, A., & Mardomingo, R. (2009). Mobile blogs in language learning: Making the most of informal and situated learning opportunities. *ReCALL*, 21(1), 96–112.
- Davis, J.F., & Lyman-Hager, M. (1997). Computers and L2 reading: Student performance, student attitudes. *Foreign Language Annals*, 30(1), 58–72.
- de Freitas, S. (2006). *Learning in immersive worlds: A review of game-based learning*. Bristol: Joint Information Systems Committee.
- Debski, R. (2002). Technology and second language learning through socialization. In S. Naidu (Ed.), *Learning and teaching with technology: Principles and practices* (pp. 129–146). Abingdon, UK: Routledge.
- deHaan, J., Reed, W.M., & Kuwada, K. (2010). The effect of interactivity with a music video game on second language vocabulary recall. *Language Learning and Technology*, 14, 74–94.
- Dodigovic, M. (2007). Artificial intelligence and second language learning: An efficient approach to error remediation. *Language Awareness*, 16, 99–113.
- Ducate, L.C., & Lomicka, L.L. (2005). Exploring the blogosphere: Use of web logs in the foreign language classroom. *Foreign Language Annals*, 38, 410–421.
- Ducate, L.C., & Lomicka, L.L. (2008). Adventures in the blogosphere: From blog readers to blog writers. *Computer Assisted Language Learning*, 21, 9–28.
- Farr, F. (2008). Evaluating the use of corpus-based instruction in a language teacher education context: Perspectives from the users. *Language Awareness*, 93(i), 61–78.
- Felix, U. (2005). Analysing recent CALL effectiveness research – Toward a common agenda. *Computer Assisted Language Learning*, 18(1–2), 1–32.
- Gettys, S., Imhof, L.A., & Kautz, J.O. (2001). Computer assisted reading: The effect of glossing format on comprehension and vocabulary retention. *Foreign Language Annals*, 34, 91–106.
- Gilgen, R. (2005). Holding the world in your hand: Creating a mobile language learning environment. *EDUCAUSE Quarterly*, 3, 30–39.
- Grace, C. (2000). Gender differences: Vocabulary retention and access to translations for beginning language learners in CALL. *Modern Language Journal*, 84, 214–224.
- Gray, L. (2008). *Effective practice with e-Portfolios*. Bristol, UK: JISC Innovation Group.
- Gray, C., Hagger-Vaughan, L., Pilkington, R., & Tomkins, S.-A. (2005). The pros and cons of interactive whiteboards in relation to the key stage 3 strategy and framework. *Language Learning Journal*, 32, 38–44.
- Hardison, D.M. (2004). Generalization of computer-assisted prosody training: Quantitative and qualitative findings. *Language Learning & Technology*, 8, 34–52.
- Harless, W.G., Zier, M.A., & Duncan, R.C. (1999). Virtual dialogues with native speakers: The evaluation of an interactive multimedia method. *CALICO Journal*, 16, 313–337.
- Heift, T., & Schulze, M. (2007). *Errors and intelligence in Computer-Assisted Language Learning: Parsers and pedagogues*. Abingdon, UK: Routledge.
- Higgins, S., Beauchamp, G., & Miller, D. (2007). Reviewing the literature on interactive whiteboards. *Learning, Media and Technology*, 32, 213–225.
- Hirata, Y. (2004). Computer assisted pronunciation training for native English speakers learning Japanese pitch and durational contrasts. *Computer Assisted Language Learning*, 17, 357–376.

- Holland, V.M., Kaplan, J.D., & Sabol, M.A. (1999). Preliminary tests of language learning in a speech-interactive graphics microworld. *CALICO Journal*, 16, 339–359.
- Hong, W. (1997). Multimedia computer-assisted reading in business Chinese. *Foreign Language Annals*, 30, 335–344.
- Hubbard, P. (2005). A review of subject characteristics in CALL research. *Computer Assisted Language Learning*, 18, 351–368.
- Ioannou, A., & Artino, A. (2008). *Incorporating Wikis in an educational technology course: Ideas, reflections and lessons learned*. Paper presented at the Society for Information Technology and Teacher Education (SITE) International Conference, Las Vegas, NV.
- Jacobs, D., & Rogers, C. (1999). Treacherous allies: Foreign language grammar checkers. *CALICO Journal*, 16, 509–531.
- Kennedy, C., & Miceli, T. (2001). An evaluation of intermediate students' approaches to corpus investigation. *Language Learning & Technology*, 5, 77–90.
- Kern, R.G. (1995). Restructuring classroom interaction with networked computers: Effects on quantity and characteristics of language production. *Modern Language Journal*, 79, 457–476.
- Kessler, G. (2009). Student-initiated attention to form in wiki-based collaborative writing. *Language Learning and Technology*, 13(1), 79–95.
- Kessler, G., & Bikowski, D. (2010). Developing collaborative autonomous learning abilities in computer mediated language learning: Attention to meaning among students in wiki space. *Computer Assisted Language Learning*, 25(1), 41–58.
- Kiernan, P.J., & Aizawa, K. (2004). Cell phones in task based learning: Are cell phones useful language learning tools? *ReCALL*, 16, 71–84.
- Kim, I.S. (2006). Automatic speech recognition: Reliability and pedagogical implications for teaching pronunciation. *Educational Technology & Society*, 9(1), 322–334.
- Kitade, K. (2000). L2 learners' discourse and SLA theories in CMC: Collaborative interaction in internet chat. *CALL*, 13, 143–166.
- Kitade, K. (2008). The role of offline metalanguage talk in asynchronous computer-mediated communication. *Language Learning & Technology*, 12(1), 64–84.
- Knight, S. (1994). Dictionary use while reading: The effects on comprehension and vocabulary acquisition for students with different verbal abilities. *Modern Language Journal*, 78, 285–299.
- Kocoglu, Z. (2008). Turkish EFL student teachers' perceptions on the role of electronic portfolios in their professional development. *Turkish Online Journal of Educational Technology*, 7. ISSN: 1303-6521.
- Kost, C., Foss, P., & Lenzini, J. (1999). Textual and pictorial glosses: Effectiveness on incidental vocabulary growth when reading in a foreign language. *Foreign Language Annals*, 32(1), 89–97.
- Koyama, T., & Takeuchi, O. (2004). Comparing electronic and printed dictionaries: How the difference affected EFL learning. *JACET Bulletin*, 38, 33–46.
- Koyama, T., & Takeuchi, O. (2007). Does look-up frequency help reading comprehension of EFL learners? Two empirical studies of electronic dictionaries. *CALICO Journal*, 25(1), 110–125.
- Kvavik, R. (2005). Convenience, communications and control: How students use technology. In D.G. Oblinger & J.L. Oblinger (Eds.), *Educating the net generation*. Boulder, CO: EDUCAUSE Center for Applied Research.
- Lai, C., & Zhao, Y. (2006). Noticing and text-based chat. *Language Learning & Technology*, 10, 102–120.
- Lan, Y.J., Sung, Y.T., & Chang, K.E. (2007). A mobile device supported peer assisted learning system for collaborative early EFL reading. *Language Learning and Technology*, 11, 130–151.
- Laufer, B., & Hill, M. (2000). What lexical information do L2 learners select in a CALL dictionary and how does it affect word retention. *Language Learning & Technology*, 3, 58–76.
- Laufer, B., & Levitsky-Aviad, T. (2006). Examining the effectiveness of "Bilingual Dictionary Plus" – A dictionary for production in a foreign language. *International Journal of Lexicography*, 19, 135–155.
- Lee, L. (2008). Focus-on-form through collaborative scaffolding in expert-to-novice online interaction. *Language Learning and Technology*, 12, 53–72.
- Lee, L. (2010). Exploring wiki-mediated collaborative writing: A case study in an elementary Spanish course. *CALICO Journal*, 27, 260–276.
- Leech, G. (1997). *Teaching and language corpora*. London: Longman.

- Leffa, V. (1993). Using an electronic dictionary to understand foreign language texts. *Trabalhos em linguística aplicada*, 21, 19–29.
- Lenders, O. (2008). Electronic glossing – Is it worth the effort? *Computer Assisted Language Learning*, 21, 457–481.
- Levy, M., & Stockwell, G. (2006). *Options and issues in computer assisted language learning*. Mahwah, NJ: Erlbaum.
- Liou, H.C. (2000). The electronic bilingual dictionary as a reading aid to EFL learners: Research findings and implications. *Computer Assisted Language Learning*, 13, 467–476.
- Little, D., & Perclova, V. (2001). *The European language portfolio: Guide for teachers and teacher trainers*. Strasbourg, France: Council of Europe.
- Liu, T.-Y. (2009). A context-aware ubiquitous learning environment for language listening and speaking. *Journal of Computer Assisted Learning*, 25, 515–527.
- Liu, D., & Jiang, P. (2009). Using a corpus-based lexicogrammatical approach to grammar instruction in EFL and ESL contexts. *Modern Language Journal*, 93(i), 61–78.
- Lomicka, L.L. (1998). “To gloss or not to gloss”: An investigation of reading comprehension. *Language Learning & Technology*, 1, 41–53.
- Loucky, J. (2005). Combining the benefits of electronic and online dictionaries with CALL web sites to produce effective and enjoyable vocabulary and language learning lessons. *Computer Assisted Language Learning*, 18, 389–416.
- Lu, M. (2008). Effectiveness of vocabulary learning via mobile phone. *Journal of Computer Assisted Learning*, 24, 515–525.
- Lund, A. (2008). Wikis: A collective approach to language production. *ReCALL*, 20, 35–54.
- Lund, A., & Smørdal, O. (2006). *Is there space for the teacher in a wiki?* Paper presented at the WikiSym '06, Odense, Denmark.
- Machovikov, A., Stolyarov, K., Chernov, M., Sinclair, I., & Machovikova, I. (2002). Computer-based training system for pronunciation. *Computer Assisted Language Learning*, 15, 201–214.
- MacWhinney, B. (1995). Evaluating foreign language tutoring systems. In V.M. Holland, J.D. Kaplan, & M.R. Sams (Eds.), *Intelligent language tutors: Theory shaping technology*. Mahwah, NJ: Lawrence Erlbaum.
- Miyazoye, T., & Anderson, T. (2010). Learning outcomes and students' perceptions of online writing: Simultaneous implementation of a forum, blog, and wiki in an EFL blended learning setting. *System*, 38, 185–199.
- Nagata, N. (1993). Intelligent computer feedback for second language instruction. *Modern Language Journal*, 77, 330–339.
- Nagata, N. (1997). An experimental comparison of deductive and inductive feedback generated by a simple parser. *System*, 25, 515–534.
- Nagata, N. (1999). The effectiveness of computer-assisted interactive glosses. *Foreign Language Annals*, 32, 469–479.
- Nagata, N. (2009). An online Japanese textbook with Natural Language Processing. In I. Lancashire (Ed.), *Teaching literature and language online* (pp. 384–409). New York: Modern Language Association of America.
- O'Brien, M.G., & Levy, R.M. (2008). Exploration through virtual reality: Encounters with the target. *Canadian Modern Language Review*, 64, 663–691.
- Orr, M. (2008). Learner perceptions of interactive whiteboards in EFL classrooms. *CALL-EJ*, 9. Retrieved June 30, 2012 from <http://callej.org/journal/9-2/orr.html>.
- Paredes, M., Sanchez-Villalon, P.P., Ortega, M., & Velazquez-Iturbide, J.A. (2007). Collaborative composition in a foreign language with handheld computing and web tools. *Journal of Universal Computer Science*, 13, 948–958.
- Payne, J.S., & Ross, B.M. (2005). Synchronous CMC, working memory, and L2 oral proficiency development. *Modern Language Journal*, 9, 35–54.
- Payne, J.S., & Whitney, P.J. (2002). Developing L2 oral proficiency through synchronous CMC: Output, working memory, and interlanguage development. *CALICO Journal*, 20, 7–32.
- Peters, E. (2007). Manipulating L2 learners' online dictionary use and its effect on L2 word retention. *Language Learning & Technology*, 11, 36–58.
- Plass, J.L., Chun, D.M., Mayer, R.E., & Leutner, D. (1998). Supporting visual and verbal learning preferences in a second-language multimedia learning environment. *Journal of Educational Psychology*, 90(1), 25–36.

- Renalli, J. (2008). Learning English with The Sims: Exploiting authentic computer simulation games for L2 learning. *Computer Assisted Language Learning*, 21, 441–455.
- Rypa, M.E., & Price, P. (1999). VILTS: A tale of two technologies. *CALICO Journal*, 16, 385–404.
- Samuels, J. (2005). *Wireless and handheld devices for language learning*. Paper presented at the 19th Annual Conference on Distance Teaching and Learning, Madison, WI.
- Sanprasert, N. (2009). The application of a course management system to enhance autonomy in learning English as a foreign language. *System*, 38(1), 109–123.
- Satar, H.M., & Özdener, N. (2008). The effects of synchronous CMC on speaking proficiency and anxiety: Text versus voice chat. *Modern Language Journal*, 92, 595–613.
- Sathe, N., & Waltje, J. (2008). The iPod project: A mobile mini-lab. *Journal of the Research Center for Educational Technology (RCET)*, 4, 32–56.
- Schmid, E.C. (2007). Enhancing performance knowledge and self-esteem in classroom language learning: The potential of the ACTIVote component of interactive whiteboard technology. *System*, 35, 119–133.
- Shekary, M., & Tahririan, M.H. (2006). Negotiation of meaning and noticing in text-based online chat. *Modern Language Journal*, 90, 557–573.
- Shih, Y.-C., & Yang, M.T. (2008). A collaborative virtual environment for situated language learning using VEC3D. *Educational Technology & Society*, 11, 56–68.
- St.John, E. (2001). A case for using a parallel corpus and concordancer for beginners of a foreign language. *Language Learning & Technology*, 5, 185–203.
- Stockwell, G. (2007a). A review of technology choice for teaching language skills and areas in the CALL literature. *ReCALL*, 19, 105–120.
- Stockwell, G. (2007b). Vocabulary on the move: Investigating an intelligent mobile phone-based vocabulary tutor. *Computer Assisted Language Learning*, 20, 365–383.
- Sullivan, N., & Pratt, E. (1996). A comparative study of two ESL writing environments: A computer-assisted classroom and a traditional oral classroom. *System*, 29, 491–501.
- Sun, Y.-C. (2009). Voice blog: An exploratory study of language learning. *Language Learning and Technology*, 13, 88–103.
- Surface, E.A., Dierdorff, E.C., & Watson, A.M. (2007). *Special operations language training software measurement of effectiveness study: Tactical Iraqi study final report*. Los Angeles, CA: Alelo.
- Sykes, J.M. (2005). Synchronous chat and pragmatic development: Effects of oral and written chat. *CALICO Journal*, 22, 399–431.
- Sykes, J.M. (2008). *A dynamic approach to social interaction: Synthetic immersive environments & Spanish pragmatics* (Doctoral dissertation). University of Minnesota, Minneapolis, MN.
- Tanner, M.W., & Landon, M.M. (2009). The effects of computer-assisted pronunciation readings on ESL learners' use of pausing, stress, intonation, and overall comprehensibility. *Language Learning & Technology*, 13, 51–65.
- Taylor, A. (2006). The effects of CALL versus traditional L1 glosses on L2 reading comprehension. *CALICO Journal*, 23, 309–318.
- Taylor, A. (2009). CALL-based versus paper-based glosses: Is there a difference in reading comprehension? *CALICO Journal*, 27(1), 147–160.
- Thorne, S.L., & Payne, J.S. (2005). Evolutionary trajectories, internet-mediated expression and language education. *CALICO Journal*, 22, 371–397.
- Thorne, S.L., Webber, D., & Bensinger, A. (2005). *Interactivity system analysis, AIM, and pedagogical innovation*. Paper presented at the CALICO conference, East Lansing, MI.
- Thornton, P., & Houser, C. (2002). M-learning: Learning in transit. In P. Lewis (Ed.), *The changing face of CALL: A Japanese perspective* (pp. 229–244). London: Taylor & Francis.
- Thornton, P., & Houser, C. (2003). *EduCall: Adding interactivity to large lecture classes in Japan via mobile phones*. Paper presented at the World Conference on Educational Multimedia, Hypermedia and Telecommunications, Honolulu, Hawaii.
- Thornton, P., & Houser, C. (2005). Using mobile phones in English education in Japan. *Journal of Computer Assisted Learning*, 21, 217–228.
- Tozcu, A. (2008). The use of interactive whiteboards in teaching non-roman scripts. *Computer Assisted Language Learning*, 21, 143–166.
- Tudini, V. (2003). Using native speakers in chat. *Language Learning & Technology*, 7, 141–159.

- Ware, P.D., & O'Dowd, R. (2008). Peer feedback on language form in telecollaboration. *Language Learning & Technology*, 12(1), 43–63.
- Warschauer, M. (1996). Comparing face-to-face and electronic communication in the second language classroom. *CALICO Journal*, 13, 7–26.
- Warschauer, M. (1997). Computer-mediated collaborative learning. *The Modern Language Journal*, 81, 470–481.
- Yeh, Y., & Wang, C. (2003). Effects of multimedia vocabulary annotations and learning styles on vocabulary learning. *CALICO Journal*, 21(1), 131–144.
- Yoon, H. (2008). More than a linguistic reference: The influence of corpus technology on L2 academic writing. *Language Learning & Technology*, 12, 31–48.
- Yoshii, M. (2006). L1 and L2 glosses: Their effects on incidental vocabulary learning. *Language Learning & Technology*, 10, 85–101.
- Yoshii, M., & Flaitz, J. (2002). Second language incidental vocabulary retention: The effect of text and picture annotation types. *CALICO Journal*, 20(1), 33–58.
- Zhao, Y. (2003). Recent developments in technology and language learning: A literature review and meta-analysis. *CALICO Journal*, 21(1), 7–28.

Appendix 1. Studies included in the evidence summary table.

Citation	Technology	Study type	Number of participants	Outcome measures
Armstrong and Retterer (2008)	Blog	Non-experimental	16	Writing assignments, oral exams, survey
Aust et al. (1993)	Electronic dictionary	Experimental	80	Frequency of consultations, study time, efficiency, proposition recall protocol for comprehension
Blake (2009)	Chat	Experimental	34	Pretest and posttest measures of fluency: speaking rate, phonation time ratio, articulation rate, mean length or run, average length of pauses
Burston (2001)	Grammar checker	Non-experimental	Semester 1 – 54; Semester 2 – 40	Analysis of student compositions and examination essays, questionnaire
M. Carey (2004) Chen (2008)	ASR Chat	Experimental Experimental	19 16	Expert judgment Vocabulary, correction, and spelling posttests (immediate and delayed)
Chen and Hsu (2008)	PDA	Non-experimental	15	Reading comprehension test, questionnaire
Chiu et al. (2007) Davis and Lyman-Hager (1997)	Intelligent tutor Electronic gloss and annotation	Non-experimental Non-experimental	49 42	Speech test, questionnaire Reading comprehension task, recall protocol, multiple-choice task, interview

(continued)

Appendix 1. (Continued).

Citation	Technology	Study type	Number of participants	Outcome measures
Dodigovic (2007)	Intelligent tutor	Experimental	266	Grammaticality judgment pre-test, short answer post-test
Ducate and Lomicka (2008)	Blog	Non-experimental	Semester 1 – 29; Semester 2 – 21	Writing assignments, focus groups, questionnaires
Gray et al. (2005)	IWB	Non-experimental	unknown	Focus meetings, teaching logs, observations, interviews, dissemination meetings
Hardison (2004)	ASR	Experimental and non-experimental	Study 1 – 26; Study 2 – 26	Study 1 – rating scale, questionnaire, observation; Study 2 – memory recall task
Harless et al. (1999)	Intelligent tutor	Non-experimental	Study 1 – 20; Study 2 – 4; Study 3 – 9	Study 1 – questionnaire; Study 2 – interviews; Study 3 – OPI, DLPT for listening and reading
Hirata (2004)	ASR	Experimental	8	Pre- and posttraining production and perception tests, expert judgment
Holland et al. (1999)	Intelligent tutor	Non-experimental	Pilot study – 6; Preliminary field trial – 16	Pretest–posttest (sentence-making skill), attitude questionnaire
Hong (1997)	Electronic gloss and annotation	Experimental	20	Conventional and computer-assisted reading comprehension tests
Ioannou and Artino (2008)	Wiki	Non-experimental	24	Observation, teacher opinion
Jacobs and Rogers (1999)	Grammar checker	Experimental	42	Error analysis
Kern (1995)	Chat	Experimental	40	Transcripts of students' writing and oral production, questionnaire

(continued)

Appendix 1. (Continued).

Citation	Technology	Study type	Number of participants	Outcome measures
Kitade (2000)	Chat	Non-experimental	11	Chat discussions, proficiency interview, questionnaire
Kitade (2008)	Bulletin board/asynchronous chat	Non-experimental	36	Individualized posttest (recall from metalinguistic episodes), retrospective questionnaire, and interview
Koyama and Takeuchi (2007)	Electronic dictionary	Experimental	Study 1 – 34, Study 2 – 31	Frequency of look-ups, time spent on tasks, reading comprehension quiz
Lai and Zhao (2006)	Chat	Non-experimental	12	Stimulated recall sessions
Lan et al. (2007)	Tablet PC	Experimental	52	Observation (frequencies of interactive behaviors)
Laufer and Levitzky-Aviad (2006)	Electronic dictionary	Non-experimental	75	Rating scale for preferences, log files
Lee (2008)	Chat	Non-experimental	30	Chat logs, reflective essay
Leffa (1993)	Electronic dictionary	Experimental	Study 1 – 20; Study 2 – 51	Study 1 – translation task, reading comprehension test; Study 2 – questionnaire
Liou (2000)	Electronic dictionary	Non-experimental	14	Reading comprehension questions, paraphrase questions, interview
Liu and Jiang (2009)	Corpus	Non-experimental	244	Corpus search projects, reflection papers, teacher lesson plans, teaching logs, teaching journals, post-study assessment survey
Loucky (2005)	Electronic dictionary	Non-experimental	39	Survey

(continued)

Appendix 1. (Continued).

Citation	Technology	Study type	Number of participants	Outcome measures
Lu (2008)	Cell phone (SMS)	Experimental	30	Pre- and post-treatment word recognition test (immediate and delayed), questionnaire, interview
Lund and Smørdal (2006)	Wiki	Non-experimental	31	Audio- and video-taped lessons, questionnaires, wiki content pages, field notes
Nagata (1993)	Intelligent tutor	Experimental	34	Achievement test, retention test, questionnaire
Nagata (1997)	Intelligent tutor	Experimental	30	Pre- and post-treatment tests measuring production of Japanese particles, retention test, comprehension test, oral test
Orr (2008)	IWB	Non-experimental	100+	Interview, open questionnaire question
Payne and Ross (2005)	Chat	Non-experimental	24	Working memory measure – reading span and nonword repetition tests. Oral proficiency measure – speaking task, speech sample, scored based on a holistic scale
Payne and Whitney (2002)	Chat	Experimental	58	Working memory measures – nonword repetition task and reading span test. Oral proficiency measure – Oral Production Interview Scale
Sanprasert (2009)	CMS	Non-experimental	55	Student journals, survey

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Appendix 1. (Continued).				
Citation	Technology	Study type	Number of participants	Outcome measures
Satar and Ozdener (2008)	Chat	Experimental	90	Prenxiety and postanxiety scales, speaking test, closed and open-ended questionnaire
Sathe and Waltje (2008) Shekary and Tahririan (2006)	iPod Chat	Non-experimental Experimental and non-experimental	More than 120 16	Survey Individualized posttest (based on language-related episodes)
Shih and Yang (2008)	Virtual world	Non-experimental	Unknown	Chat log and event history analyzed by using ethnographic approach, questionnaire
Sullivan and Pratt (1996)	Chat	Experimental	38	Writing apprehension scale, attitudes toward writing with the computer scale, writing samples
Surface et al. (2007)	Serious games	Experimental	421	Pretest, mid-training, and posttest questionnaires. Posttest knowledge assessment. Data logs of training performance
Taylor (2006)	Electronic gloss and annotation	Meta-analysis	n/a	n/a
Taylor (2009)	Electronic gloss and annotation	Meta-analysis	n/a	n/a
Thornton and Houser (2005)	Cell phone (SMS)	Experimental	Study 1 – 13; 68	Pretest and posttests of vocabulary, questionnaire
Tozcu (2008)	IWB	Non-experimental	75	Questionnaire

(continued)

Appendix 1. (Continued).

Citation	Technology	Study type	Number of participants	Outcome measures
Warschauer (1996)	Chat	Experimental	16	Questionnaire, qualitative content analysis. Language complexity measures – type-token ratio, coordination index
Yoon (2008)	Corpus	Non-experimental	6	Classroom observation notes, interviews, recall protocols, corpus search logs, class corpus search assignments, written reflections